

Dynamic Routing and Scheduling of Mobile Charging Stations for Electric Vehicles Using Deep Reinforcement Learning

Ubaid Qureshi, *Student Member, IEEE*, Arnob Ghosh, *Member, IEEE* and B K Panigrahi, *Fellow, IEEE*

Abstract—This paper presents an innovative solution for charging electric vehicles (EVs) on the go. Unlike traditional charging stations, our proposed system schedules and routes mobile charging stations (MCSs) to provide charging services to EVs at their preferred location and time. However, the dynamic and evolving nature of EV charging requests requires a real-time approach to optimize the scheduling and routing of MCSs. To address this challenge, we propose a distributed model-free deep reinforcement learning approach for the dynamic routing of MCSs. The MCSs learn the optimal policy by interacting with the environment in a distributed manner without explicitly needing to model the system. Numerical results demonstrate that our approach provides optimal charging solutions to meet the growing demand for EV charging.

Index Terms—scheduling, routing, electric vehicle charging, deep learning, mobile charging stations, model free learning.

I. INTRODUCTION

A. Motivation

ELECTRIC Electric vehicles are expected to increase leading up to 2030 and are thus going to present new challenges and opportunities. The biggest challenge in the large-scale transition from internal combustion engine (IC) based vehicles to electric vehicles (EV) will be the requirement for charging infrastructure. As the time to charge an electric vehicle (30 minutes) is considerably longer than the time needed to refill the IC engine vehicles, this will lead to congestion and significant waiting times at commercial charging stations as EV penetration increases. In addition, the EV owners will face the risk of range anxiety about whether they can complete a trip given the charge left in their batteries. Lastly, the uncontrolled charging of EVs at peak load times will worsen the load curve and increase the need for power system augmentation.

We propose a mobile charging service for electric vehicles using vans equipped with fast level-3 DC-DC chargers and a battery bank. EV owners request charging via a desktop or mobile client, and the charging service operator assigns the requests to the minimum number of mobile charging stations. The vans need to be scheduled and routed to EV charging requests and also recharge at fixed charging stations.

U. B. Qureshi is with the Department of Electrical Engineering, Indian Institute of Technology, Delhi, 110016 India, and also with the University of Kashmir, Srinagar, 190006, India. (email: ubaidqureshi4209@gmail.com)

A. Ghosh is with the Department of Electrical and Computer Engineering, New Jersey Institute of Technology, Newark, New Jersey 07102 USA.

B. K. Panigrahi is with the Department of Electrical Engineering, Indian Institute of Technology, Delhi, 110016, India.

The proposed scheme of charging has three main advantages:

1) EV owners will not have to spare additional time to take their electric vehicles for charging, thus solving the congestion problem at public charging stations. 2) The EVs can be charged without location and time constraints, thus solving the problem of range anxiety. 3) The proposed mobile charging service decouples load from the grid on a real-time basis and thus helps flatten the load curve.

To the best of our knowledge this is the first attempt to address the problem of matching MCSs to EV requests using deep reinforcement learning. However, there have been attempts to enhance the available charging facilities through various papers, such as [6] - [9]. Mobile charging services have the potential to solve almost all problems in large-scale electric vehicle charging [1], [10]. The biggest challenge for mobile charging services is to service maximum EV requests using a minimum number of MCSs. This requires scheduling multiple requests to a single MCS, but the electric vehicle charging requests are random [1]. Therefore, there needs to be a way to control the charging requests to match the supply and demand of mobile charging stations.

The literature on electric vehicle charging infrastructure highlights several key themes. One of the most important is the need for reliable and convenient charging options, which has been discussed in multiple studies [10] - [12]. Another theme is the impact of charging station accessibility on electric vehicle adoption, which has been studied in research papers like [4], [7] - [10]. Technical and operational requirements for mobile charging stations have also been explored in literature [1] - [5]. Finally, the economic and environmental benefits of widespread deployment of electric vehicle charging infrastructure have been studied extensively. Mobile charging stations can play a vital role in expanding the reach of electric vehicle charging infrastructure, especially in urban areas where space for traditional charging stations may not be feasible or economical [2]. However, mobile charging stations also face several challenges, including limited battery capacity [11] and the need for a sufficient power supply [12].

B. Problem Statement

Mobile Charging Service Scenario: For example, figure 1 has seven charging requests from non-overlapping regions A, C, D, E, F, G, and H and two mobile charging stations, MCS 1 and MCS 2. The two MCSs must recharge their battery bank at either fixed charging stations at C and F. There are

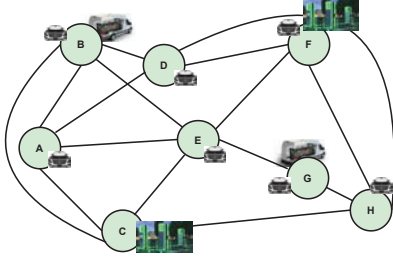


Fig. 1. Schematic to demonstrate MCS in a particular area.

several possible ways for the two mobile charging stations to service the EV charging requests, such as MCS 1 servicing all the charging requests in a specific sequence or MCS 1 servicing some charging requests while MCS 2 services others. Our proposed solution uses distributed deep reinforcement learning to design a mobile charging service that leverages the benefits of pooling mobile charging stations to service multiple EV charging requests while overcoming the challenges of accurately modeling the system.

Thus each mobile charging station decides in a distributed manner which zone it should go next to service scheduled or anticipated EV charging requests while ensuring to recharge its battery bank in one of the fixed charging stations without having to communicate with other MCSs. The routing and scheduling should be such as using the minimum number of mobile charging stations while maximizing the operating agency's profit.

C. Contribution

The service area is divided into zones, represented by the set $i = \{1, 2, 3, \dots, N\}$. Mobile charging stations are denoted by the set M . The total scheduling horizon T is divided into small, equal time intervals ΔT . The goal is to optimize the system for the entire scheduling horizon by assigning charging requests from various zones to different mobile charging stations M in real-time at each interval ΔT . At any given time, let $m_{t,n}$ represent the number of mobile charging stations available in zone n . Because mobile charging stations have limited charging capacity, they can only charge a specific number of EVs, which is represented by Z_{max} . A mobile charging station can only service another charging request if it has energy left in its battery bank and is willing to do so. Servicing additional requests may delay the time the already scheduled request will be serviced, which is not ideal. Similarly, the number of charging requests in zone n at time t is $E_{t,n}$.

Next, we define the state vector. The state of environment at any instant of time can be represented by the tuple (X_t, V_T, D_T) , where X_t represents the state of mobile charging stations; *present location of MCS, Energy left in MCS to take new requests, location of scheduled charging requests, time at which the scheduled request finishes, the MCSs that are already in the central charging station*. V_T represents the predicted mobile charging stations at each zone at various intervals of time and D_T represents the estimated demand at various zones at various intervals of time. Though

not known with certainty, it is possible to predict the demand at various zones for different time intervals from previous charging request patterns. In summary the contributions of our work are as follows:

- 1) We formalize the mobile charging service considering stochastic nature of charging requests and travel times of mobile charging stations on transportation network.
- 2) We formulate a multi-objective optimization problem where the dispatcher simultaneously want to minimize five objectives:-i) the number of mobile charging stations, ii) the travelling time of mobile charging stations, iii) the miss ration of charging requests, iv) the breach of charging deadlines and v) the maximum demand and congestion at fixed charging station.
- 3) We use distributed model-free deep reinforcement learning to optimize real-time joint scheduling and routing of mobile charging stations under stochasticity.

II. PROPOSED FRAMEWORK DESIGN

A. Objective

We present the overall central objective function that the charging station operator aims to minimize. We explain each of the metric in detail below. The proposed DeepRouting framework is designed to optimize a number of objectives. The decision variables are: i) $u_{t,j}^m = 1$ if vehicle m should be dispatched to zone j for time interval t . ii) $\zeta_{t,n}^m = 1$ if the MCS m decides to serve the EV charging requests at zone n during time interval t , otherwise $\zeta_{t,n}^m = 0$. iii) $\psi_{t,m} = 1$ if the MCS m decides to recharge its battery at a fixed charging station during time interval t otherwise $\psi_{t,m} = 0$. The various objectives of the proposed DeepRouting model are:

1) Minimizing the number of mobile charging stations:

We define a metric which measures the number of additional mobile charging stations that are operated in the present time interval over the preceding time interval.

$$M_t = \sum_{n=1}^N [\max(m_{t,n} - m_{t-1,n}, 0)] \quad (1)$$

where $m_{t,n}$ is the number of mobile charging stations in zone n during time interval t and $m_{t-1,n}$ is the number of mobile charging stations active in zone n during the preceding time interval $t - 1$. The goal of routing policy is to minimize the metric M_t for each interval of time.

2) *Minimizing the travelling time of mobile charging stations:* We define the metric which measures the total travel time of mobile charging stations during each interval of time.

$$T_t = \sum_{m=1}^M \sum_{j=1}^N d_{t,j}^m u_{t,j}^m \quad (2)$$

where $d_{t,j}^m$ is the estimated distance the mobile charging station m has to travel if it is dispatched during interval t to zone j and the $u_{t,j}^m$ is the decision variable, $u_{t,j}^m = 1$ if the m^{th} mobile charging station is dispatched to zone j during time interval t .

3) *Minimizing the miss ratio of charging requests:* The miss ratio or demand-supply mismatch should be as small as

possible. The metric that measures the miss-ratio of charging requests is defined as:

$$S_t = \sum_{n=1}^N \max[(\bar{z}_{t,n} - a_{t,n}), 0] \quad (3)$$

where $\bar{z}_{t,n}$ represents the estimated number of EV charging requests at zone n during time interval t and $a_{t,n}$ represents the number of active MCSs at zone n during time interval t . $a_{t,n}$ depends on the number of mobile charging stations m in that zone and the energy left in the mobile charging stations. $a_{t,n} = \sum_{m=1}^M \zeta_{t,n}^m \mathbb{1}(MCS\ m\ is\ at\ zone\ n)$ where $\zeta_{t,n}^m$ is the binary decision variable that determines if the m^{th} mobile charging station decides to service EV charging request at zone n during time interval t .

4) *Penalizing the missing of charging deadline* : We introduce the following metric:

$$\Delta_t = \sum_{n=1}^N \sum_{l=1}^L \max[(t - d_{n,l}), 0] \quad (4)$$

where L is the number of active EV charging requests in zone n and time interval t and $d_{n,l}$ represents the deadline of charging requests l in zone n .

5) *Reducing the maximum demand on the fixed charging stations*: The MCSs need to be recharged at periodic intervals at the fixed charging stations which are present at specific locations. However the MCSs should not come at the fixed charging stations all at once which will result in peak load on distribution system and unnecessary waiting time. Let $K \subset N$ be the set of nodes with fixed charging station. We thus introduce the following metric:

$$C_t = \sum_{m=1}^M \sum_{k=1}^K |u_{t,k}^m \psi_{t,m}|^2 \quad (5)$$

where $u_{t,k}^m$ is a control variable which denotes MCS m is available in zone k at time interval t and $\psi_{t,m} = 1$ if MCS m decides to charge its battery at time interval t .

6) *Ensuring MCS's does not run out of charge*: In order to ensure that MCS's do not run out of charge while serving the EV charging requests, we add one more metric:

$$V_t = \sum_{m=1}^M (e_{max}^m - e_{t,m})^2 \quad (6)$$

where E_{max}^m is the maximum battery capacity of MCS and $E_{t,m}$ is the battery charge level of MCS m at time interval t .

The overall objective function becomes:

$$\mathcal{P} = -[\gamma_1 M_t + \gamma_2 T_t + \gamma_3 S_t + \gamma_4 \Delta_t + \gamma_5 C_t + \gamma_6 V_t] \quad (7)$$

where $\gamma_1, \gamma_2, \gamma_3, \gamma_4, \gamma_5$ and γ_6 are the penalty factors corresponding to different objectives and negative sign indicates that the function needs to be minimized.

B. DeepRouting Framework

The state vector containing the current location of the mobile charging scheme, its remaining energy, scheduled service requests, start and end times of service requests, the availability of MCS in different zones based on the current routing and scheduling policy, and anticipated demand in

various zones is monitored by the operator. The DQN agent's objective is to learn the best policy by interacting with the environment, and taking actions that maximize rewards over time. Thus, the DQN agent seeks to learn optimal policies by continuously interacting with the environment. The three fundamental principles of reinforcement learning are:

1) *State*: The environment can be represented by the state which includes the status of all the mobile charging stations (agents), the availability of mobile charging stations at various zones for future intervals, and the expected demand at different zones for future time intervals. The state vector consists of three tuples: $S_t = [X_t, V_{t:T}, D_{t:T}]$. X_t includes the present location of the MCV, the remaining energy in MCV, the location of already scheduled EV requests for MCV, the start time and end time of the requests. The availability of active MCVs at different locations from the current time to a specific time slot ahead is represented by the state $V_{t:T}$. Depending on the dispatch decisions, it's possible to know the state vector $V_{t:T}$ accurately. The expected EV charging requests at various zones for different time intervals are represented by the state $D_{t:T}$. Whenever a new EV charging request is received, the charging station operator updates the state vector to reflect the change in the environment. The DQN engine takes the three-tuple state vector as input to decide the routing policy.

2) *Action*: The action taken by the MCV in each state is based on its state vector, which includes the MCV's location and the present location of all other MCVs, without considering their future locations. Consequently, the MCV's action is shortsighted with regard to the future decisions and locations of other MCVs. The MCV agents take action independently without coordination with one another and in a distributed manner. In each state, the MCV agent takes action, advances to the next state, and receives a reward simultaneously. The MCV agent's action space includes three variables: (1) $u_{t,j}^n$, which denotes the zone j that MCV n will service at time interval t ; (2) $\zeta_{t,n}^m$, a binary decision variable that indicates whether MCV n will service EV requests during time interval t ; and (3) $\psi_{t,m}$, another binary decision variable that indicates whether MCV m will recharge its battery at a fixed charging station.

3) *Reward*: We will now discuss the reward function for the mobile charging vehicles (MCVs) received after taking action in a particular state. In the case of the distributed policy DQN, each MCV m takes its reward independently at time t depending on its action at that interval of time. The reward function is similar to the overall objective function stated in equation 7. The reward $r_{t,m}$ of the m^{th} agent during time interval t depends on the number of EV requests being serviced by that mobile charging station ($b_{t,m}$). The goal is to maximize the number of requests serviced by the agents. The metric that measures the number of MCVs that need to be used is $M_{t,m}$. $T_{t,m}$ is the metric that corresponds to the travelling times and fuel costs of MCVs, corresponding to the number of mobile charging stations used. The metric that measures the mass ratio of charging requests is $S_{t,m}$. The metric that measures the missing of charging deadline is $\Delta_{t,m}$. $C_{t,m}$ is a

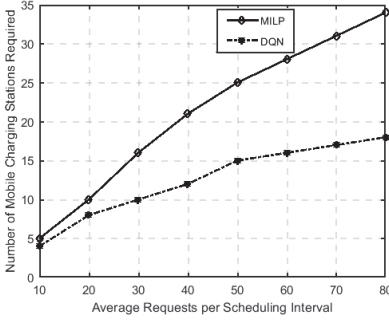


Fig. 2. Comparison of MCSs needed in DQN policy versus MILP formulation.

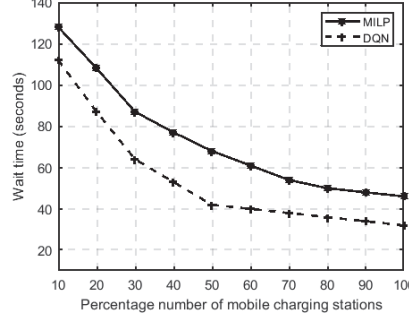


Fig. 3. Average wait time of EVs above their preferred charge beginning times.

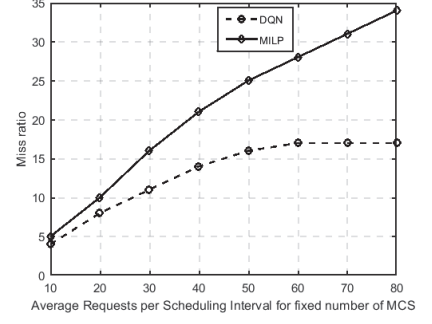


Fig. 4. Comparison of miss ratio of EV charging requests in two policies.

metric to measure the load on the fixed charging stations. V_t ensures that MCVs don't run out of charge while servicing requests. The overall discounted future reward for the MCSs is expressed as:

$$\mathcal{R}_{t,m} = \sum_{f=t}^{\infty} \eta^{f-t} r_{k,m} \quad (8)$$

where $\eta < 1$ is the discount factor. To learn the optimal policy π^* for the Reinforcement learning agent, we use Q-learning. Since the function is not explicitly defined in terms of actions, we need to learn the probabilistic dependence between the two. Given the large number of states in our case, it's impossible to use a Q-table to keep track of the rewards. Hence, we propose using convolutional neural networks to learn the transition probability and rewards from a particular state. Specifically, we are using model-free DQN to learn the probability that MCS m goes to state S_{t+1} from state S_t after taking action $a_{t,m}$ at time interval t

$$\mathcal{P}(S_{t+1}|S_t, a_{t,m}) \quad (9)$$

C. Deep Q-Network

We use deep Q-networks to generate optimized values for dispatch actions in different states of various MCSs. This learning technique is widely used in modern decision-making due to its adaptability to dynamic features in the system. The maximum expected achievable reward defines the optimal action-value function for vehicle n .

$$Q^*(s, a) = \max_{\pi} E \left[\sum_{k=t}^{\infty} \eta^{k-t} r_{k,n} \right] \quad (10)$$

The dispatcher monitors the current state and feeds it to a neural network to generate an action at any given time slot t . To determine the expectation, we do not use a complete representation of S_t . Instead, we use a neural network to approximate the Q function. For each vehicle n , an action is taken to maximize the neural network output. Initially, the learning starts with zero knowledge, and activities are chosen using a greedy scheme by following the Epsilon-Greedy method. Under this policy, the agent selects the action that results in the highest Q-value with probability $1 - \epsilon$ if not, it chooses a random action.

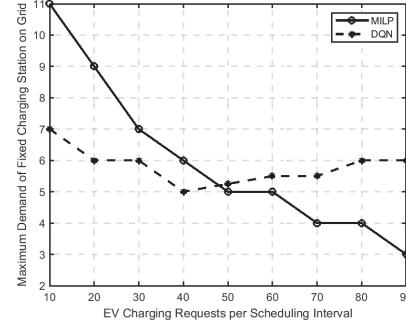


Fig. 5. Comparison of maximum demand on fixed charging station in two policies.

III. SIMULATION RESULTS

To comprehensively assess the effectiveness of our DQN policy, we conducted simulations within a service area of 400 km^2 , dividing it into small 1 km^2 elements. Charging requests were randomly generated across these area elements to emulate real-world scenarios. Our comparative analysis juxtaposes the outcomes of the DQN policy against those obtained through solving MILP, focusing on key performance parameters.

1) *Number of MCSs needed:* The fundamental metric of system efficiency, the number of required Mobile Charging Stations (MCSs), is significantly lower in the proposed DQN policy compared to the MILP formulation, as illustrated in Figure 2. This reduction implies optimal resource allocation, emphasizing the practical advantages of our DQN policy.

2) *Wait time of charging requests:* Figure 3 portrays the wait time experienced by EVs, representing the duration between the scheduled and actual start of charging. Our DQN policy demonstrates reduced wait times compared to the MILP formulation, indicating improved service efficiency and timely charging initiation.

3) *Miss ratio:* The miss ratio, quantifying the percentage of EV charging requests that could not be accommodated due to infeasibility, is a critical metric. Figure 4 showcases a significantly lower miss ratio in the DQN policy, emphasizing its superior ability to handle dynamic charging requests compared to the MILP approach.

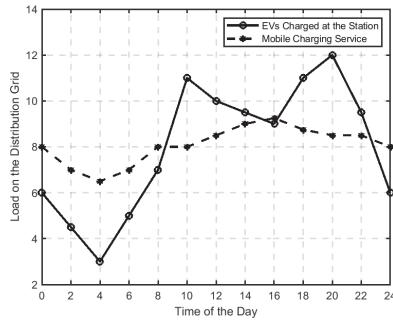


Fig. 6. Comparison of maximum demand on the distribution grid in the mobile charging service versus the direct EV charging at charging station

4) *Maximum demand on charging stations:* Figure 5 shows the maximum demand for fixed charging stations under two different scheduling policies. The DQN policy leads to a more balanced distribution of energy, preventing excessive load peaks and contributing to the overall stability of the grid. Additionally, Figure 6 compares EVs charged directly at commercial charging stations versus those charged using MCS. By using real-time decoupling of EV load from the grid and intelligent scheduling, MCS significantly improves the load profile.

A. Discussion

The observed improvements in the number of MCSs needed, wait times, miss ratios, and charging station load distribution underscore the practical advantages of our DQN policy. By adapting dynamically to changing conditions, our approach optimally allocates resources, ensuring efficient EV charging services. The DQN policy's ability to flatten the load curve, as evident in the reduced peaks in the maximum demand on charging stations, aligns with the objective of decoupling the EV load from the grid. This has positive implications for grid stability and contributes to a more balanced and sustainable energy distribution.

In summary, our simulation results not only validate the effectiveness of the DQN policy in optimizing the mobile EV charging service but also provide valuable insights into its impact on grid load management. The observed improvements establish the proposed approach as a promising solution for addressing the challenges associated with the growing demand for electric vehicle charging infrastructure.

IV. CONCLUSION

In conclusion, our study introduces a dynamic mobile charging service for electric vehicles (EVs), leveraging Distributed Model-Free Deep Reinforcement Learning (DQN). Through rigorous simulations, we showcase the efficiency of the DQN policy in reducing the number of required Mobile Charging Stations (MCSs), minimizing wait times for EV charging requests, and improving overall service accessibility. Notably, our approach contributes to load curve flattening, indicating a positive impact on grid stability. The findings underscore the practical feasibility and advantages of DQN-based scheduling, offering a promising solution to the evolving challenges in the electric vehicle charging landscape.

Future directions for this research involve transitioning from simulations to real-world implementations, considering geographical variations, traffic dynamics, and energy pricing models. Continuous optimization and adaptation of the DQN policy through ongoing learning will be crucial to enhance its performance in dynamic environments. Additionally, exploring the integration of renewable energy sources, advanced forecasting techniques, and incorporating user preferences can further contribute to the long-term sustainability and scalability of the proposed mobile charging service. The intersection of artificial intelligence, smart infrastructure, and sustainable energy solutions presents an exciting avenue for shaping the future of electric vehicle charging.

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